

Lunar geochemistry from X-ray line flux ratios using CLASS on Chandrayaan 2

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ABSTRACT

Global lunar chemical maps are essential for understanding the origin and evolution of the Moon, its surface characteristics, and its potential for resource extraction. While lunar elemental abundances have been successfully derived using model-dependent X-ray spectroscopy, these methods often require concurrent high-quality solar monitoring and intensive computational modeling, which can limit spatial coverage. Here we used X-ray fluorescence line intensity of O, Mg, Al, Si, Ca and Fe derived from five years of data from the Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS) to generate global O/Si, Mg/Si, Al/Si, Mg/Al, Ca/Si and Fe/Si line intensity ratio maps at a resolution of 5.3 km/pixel. We have developed an independent data analysis methodology for CLASS, based on open source Python packages. Our analysis shows that Mg/Al and Al/Si line intensity ratios serve as effective empirical proxies for major geochemical terranes, showing good correlation with published elemental maps, returned sample data, and the Apollo 15 and 16 X-ray Fluorescence Spectrometer (XRS) maps. Further, we apply Gaussian mixture models to the Mg/Si vs Al/Si density maps to map geochemically distinct regions on the Moon that could be of interest for future investigations.

Keywords: Lunar composition, X-ray fluorescence spectroscopy, Chandrayaan-2

1. INTRODUCTION

The lunar surface has been classified into three main crustal terranes (Jolliff et al. 2000) based on global surface geochemistry. These three large units, namely the Procellarum KREEP Terrane (PKT), Feldspathic Highlands Terrane (FHT), and the South Pole–Aitken Terrane (SPAT) (Figure 9) have uniquely different evolutionary histories. Although quantitative elemental abundance maps, such as those obtained from X-ray and gamma-ray spectroscopy, are necessary to model

the evolution of the distinct compositions, their usage has been mainly limited to these broad terranes because of the coarse spatial resolutions of such lunar elemental maps. Future lunar missions that explore a small region of interest and sample returns would require more detailed mapping to identify potential target sites.

X-ray fluorescence (XRF) spectroscopy is a well established technique in planetary remote sensing of airless bodies (Adler et al. 1972; Adler et al. 1972; Lim & Nittler 2009; Swinyard et al. 2009; Narendranath et al. 2011; Nittler et al. 2011; Weider et al. 2014; Dong et al. 2016) for elemental mapping. XRF is triggered when high-energy photons ionize inner-shell electrons, causing the emission of X-ray fluorescent photons unique to each element. The intensity of the element's spectral lines is

proportional to its abundance. However, determining elemental abundances using the XRF spectra depends on several factors, such as the solar spectrum, the geometry of the observation, the grain size, the matrix of elements present, and their possible vertical layering (Shiraiwa & Fujino 1966; Clark 1979; Clark & Trombka 1997; Okada & Kuwada 1997; Maruyama et al. 2008; Ogawa et al. 2008; Näränen et al. 2008, 2009; Weider et al. 2011; Athiray et al. 2013; Athiray et al. 2013). The uncertainties in each of these factors are propagated to the elemental abundance maps, which limits their accuracy.

Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS; Radhakrishna et al. 2020) measures lunar X-ray spectra with a nominal spatial resolution of $12.5 \text{ km} \times 12.5 \text{ km}$ from a circular polar orbit of $100 \pm 20 \text{ km}$. The lunar X-ray line flux is strongly dependent on solar activity, and often several spectra spanning larger ground pixel tracks have to be co-added to increase statistics. Narendranath et al. (2024) derived lunar elemental maps at a spatial resolution of $150 \text{ km} \times 12.5 \text{ km}$ from CLASS. Although spectral measurements were nearly global, abundance maps were limited in coverage as only the best spectral fits with good statistics could be used.

X-ray line flux ratios are representative of geochemistry, as shown by XRF experiments, such as lunar measurements on Apollo 15 (Adler et al. 1972) and 16 (Adler et al. 1972), as well as measurements for Asteroid 433 Eros (Lim & Nittler 2009) and Mercury (Weider et al. 2015; Nittler et al. 2020).

Here, we used five years of data from CLASS to generate global elemental line intensity maps of O/Si, Mg/Si, Al/Si, Mg/Al, Ca/Si and Fe/Si. Using partially overlapping tracks from multiple passes, we bin data into a uniform grid of $5.3 \text{ km} \times 5.3 \text{ km}$, which is higher than the inherent spatial resolution of CLASS.

This paper is organized as follows. In §2, we discuss the new independent pipeline that we developed for analysis of the CLASS L1 data. The ratio maps are presented in Section §3 followed by a discussion in §4. The source data was downloaded from the ISRO Science Data Archive¹, and our codes and data products are publicly available on GitHub² and Zenodo³.

2. METHODOLOGY

CLASS L1 data consist of energy-calibrated lunar X-ray spectra at a cadence of 8 s, which corresponds to a

ground pixel of $12.5 \text{ km} \times 12.5 \text{ km}$ on the lunar surface from an altitude of 100 km. We used L1 data from November 2019 to May 2025 for our analysis. Although the total number of spectra is close to 7 million, not all of them have a lunar XRF signal due to the dependence on solar activity. We therefore generated algorithms for filtering out lunar X-ray spectra where XRF signals are detected. Although the XRF signals are line emissions, there is a continuum arising from the charged particles in the lunar environment as well as scattered X-rays. The appropriate background has to be subtracted from the spectra to confirm the detection of XRF lines and ensure that we obtain accurate line ratios.

2.1. Background Estimation

The background in CLASS detectors arises from Galactic cosmic rays, solar energetic particles, and geotail particles. We estimate this charged particle background from the night-side observations. CLASS mainly collects data on the sunlit portion of the Moon, which we refer to as **light** data. It also collects about five minutes of data on the night-side of the moon, which we refer to as **dark** data, and it can help in estimating the instrument background. We note that the orbiter is not fully in the Moon's shadow throughout the **dark** phase, because of its altitude h above the lunar surface, which can lead to improper modeling of the background. For example, we note the presence of a strong Al-K α and a weak Cu-K α line in the **dark** spectra at the time of transition from **light** to **dark**. These lines are excited from the optical light-blocking filters as solar X-rays illuminate the filters and the copper collimator at grazing incidence, before the orbiter is fully occulted by the Moon. We filter the **dark** data using the following condition:

$$R_M(\csc \theta - 1) \geq h, \quad (1)$$

where R_M is the radius of the Moon, θ is the solar angle (the angle between the surface normal and sun vector), and h is the orbiter altitude given in the FITS data headers. In addition, **dark** spectra with strong instrumental lines were also filtered out.

We use the spectra measured in this filtered subset of **dark** spectra and subtract it from the **light** spectra before estimating elemental abundances. Since the background in the **light** and **dark** stages will not be exactly identical due to the missing direct solar contribution, we apply a first-order correction for this as discussed in §2.2. The Al-K α and a weak Cu-K α seen at the start and end of the **dark** data are not expected to be present in the **light** data as sun rays cannot directly reach the optical filters due to the spacecraft orientation.

We also note that the amount of **dark** data per lunar day that is occulted is a very small fraction ($\approx 14\%$)

¹ <https://pradan.issdc.gov.in/ch2/>

² <https://github.com/ravioli1369/selenomapper.git>

³ <https://doi.org/10.5281/zenodo.16726150>

of the total **dark** data, and there are often no occulted data for some orbits. We take a dynamic yearlong phase-wise background (Figure 1), where we define the phase as ± 3.5 days around the Full Moon (FM), New Moon (NM), First Quarter (FQ) and Third Quarter (TQ) to account for the phase-dependent background variations. The pipeline calculates and stores background spectra for each occurrence of these phases. When analyzing **light** data for abundances, the pipeline uses the background measurements from a period of ± 6 months from the Earth day of the **light** observation. Sample background spectra for the four phases for 2020 are shown in Figure 1.

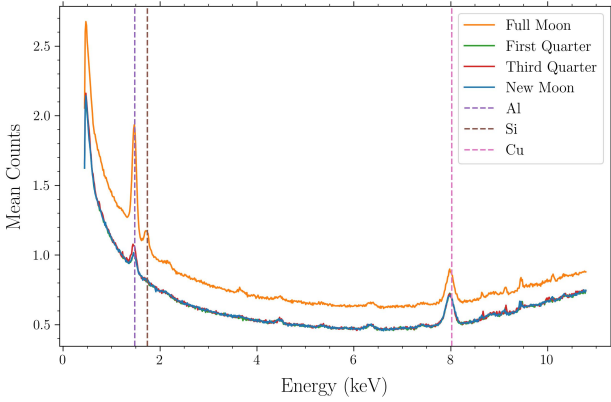


Figure 1. Mean background spectrum from one year (2020) of CLASS data measured in the night side of the orbit at different lunar phases is shown. During geotail passes, the mean background is higher with strong instrumental lines of Al-K α , Cu-K α , and a weak Si-K α .

2.2. XRF detection and analysis

Based on the energy range of CLASS, we analyze spectra to search for XRF lines of Oxygen-K α (0.52 keV), Sodium-K α (1.04 keV), Magnesium-K α (1.25 keV), Aluminum-K α (1.48 keV), Silicon-K α (1.74 keV), Calcium-K α (3.69 keV) and β (4.01 keV), Titanium-K α (4.51 keV), Chromium-K α (5.41 keV), Manganese-K α (5.89 keV) and Iron-K α (6.4 keV) and L (0.7 keV).

We estimate the detection threshold based on the variation in background counts (from the set of selected **dark** spectra) over time at the energy of each line. Given the energy resolution of CLASS, widths of typical lines, and the spacing between the lines listed above, we opted to measure background counts from a 0.25 keV window centered at each line energy. Consider a transition line T for some element X with an energy E_T^X . The total background counts in the $E_T^X \pm 0.125$ keV window at a lunar phase P are denoted by $\text{Counts}_T^X(\mathcal{B}_P)$,

and the peak amplitude within that window is denoted by $\text{Amp}_T^X(\mathcal{B}_P)$. Using the year-long set of dark measurements, we now estimate the standard deviations of the counts, $\sigma(\text{Counts}_T^X(\mathcal{B}_P))$; and of the amplitude, $\sigma(\text{Amp}_T^X(\mathcal{B}_P))$, which are used in detection criteria.

We observe the standard deviation of the background Counts during different moon phases follows the trend $\sigma(\mathcal{B}_{FM}) \gg \sigma(\mathcal{B}_{FQ}) \approx \sigma(\mathcal{B}_{TQ}) > \sigma(\mathcal{B}_{NM})$, further reinforcing the choice of using a phase-dependent background.

The instrument acquires data in 8 s exposures. Following Narendranath et al. (2024), we group the data into 96 s bins for our analysis. Defining $\mathcal{L}_t$ as the **light** data corresponding to the bin starting at time t , we obtain the integrated counts in each line T as $\text{Counts}_T^L(\mathcal{L}_t)$ using a similar 0.25 keV window centered on the line. We have to correct this for two effects. The first is simply the appropriate background for the corresponding line in that lunar phase $\mathcal{B}_P$. The second is the difference between the **light** and **dark** side spectra due to the scattered solar X-rays or particle interactions in the **light** data, discussed in §2.1. We first detrend the **light** observations by subtracting the corresponding phase-wise **dark** spectra ($\mathcal{B}_P$). Empirically, the resulting residual spectra ($\mathcal{L}_t - \mathcal{B}_P$) exhibit a relatively flat baseline but may retain a minor constant count offset. This residual offset is cleared by calculating the median counts in the line-free 2–7 keV region and subtracting this value from the entire spectrum:

$$\mathcal{L}_t^s = (\mathcal{L}_t - \mathcal{B}_P) - \text{median}(\mathcal{L}_t - \mathcal{B}_P)_{2-7 \text{ keV}}. \quad (2)$$

We now look for statistically significant detections of fluorescence lines in the normalized spectra, defined as

$$\mathcal{N}_t = \frac{\text{Counts}_T^X(\mathcal{L}_t^s)}{\sigma(\text{Counts}_T^X(\mathcal{B}_P))} \quad (3)$$

where the subscript t symbolizes the time stamp of observation during that orbit. Nominally, a 5- σ detection here would be defined as $\mathcal{N}_t > 5$. However, since the background measurement is for a long duration, it can have significant variations, increasing the denominator in Equation 3. As a consequence, this threshold may not be crossed in periods of low solar activity (Figure 3). In such periods, we apply a different threshold based on the median and standard deviation of the light curve of the line flux during that orbit, given by:

$$\mathcal{N}_t \geq \text{median}(\mathcal{N}_t) + 5\sigma(\mathcal{N}_t), \quad (4)$$

Thus, the final detection criterion applied for each element is as follows:

$$\mathcal{N}_t \geq \min(5, \text{median}(\mathcal{N}_t) + 5\sigma(\mathcal{N}_t)) \quad (5)$$

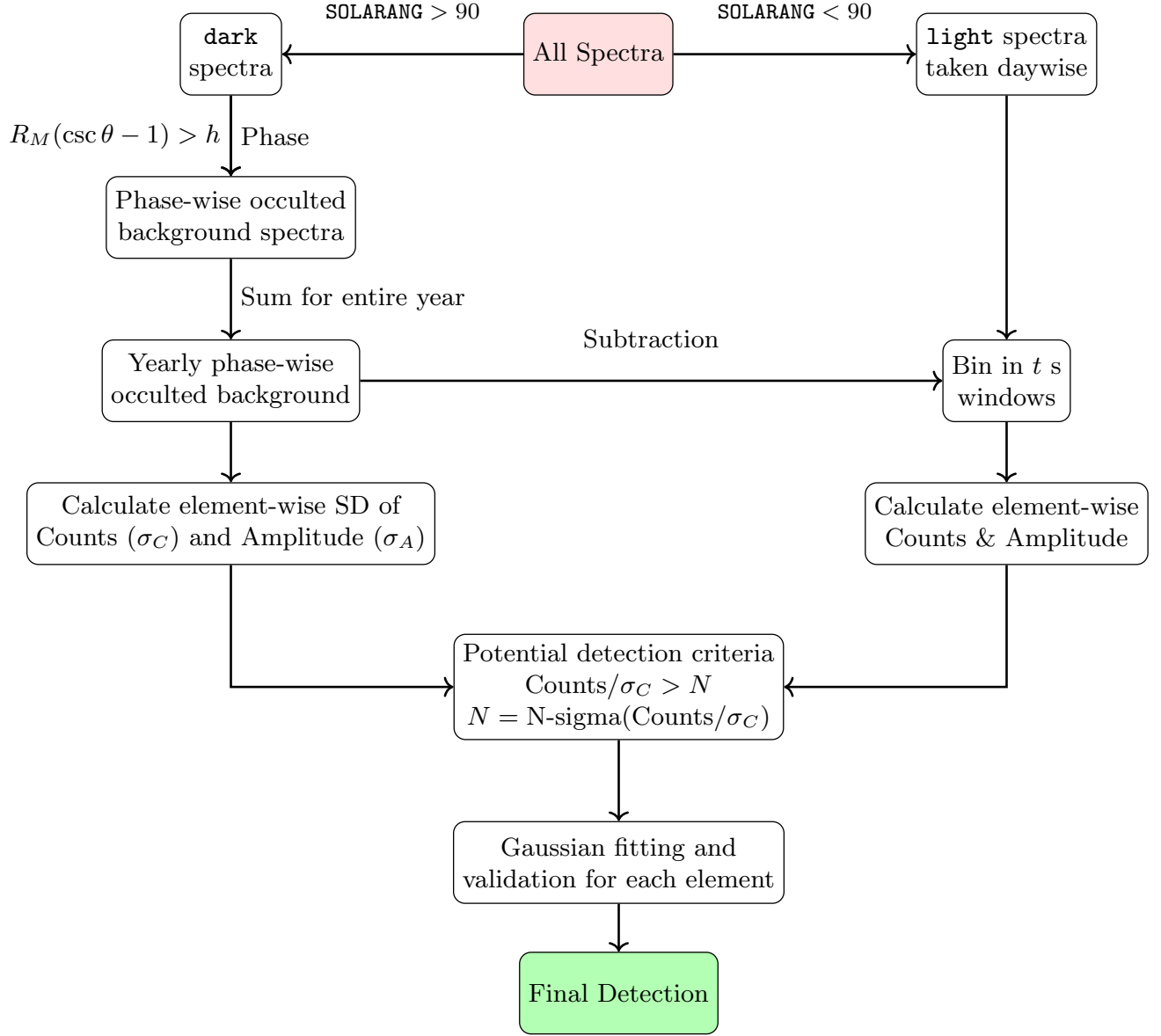


Figure 2. Flow chart of detection algorithm

Typical light curves with the N -sigma, and standard 5-sigma detection thresholds are shown in Figure 3. The N -sigma threshold defined by Equation 4 identifies far more detections during periods of low solar activity compared to the 5-sigma threshold.

Data satisfying Equation 5 are now analysed to measure line strengths. We fit the spectral lines with a Gaussian function ($\mathcal{G}$) to obtain the integrated counts under each line. We use `specutils`'s `fit_lines` function for fitting Astropy's `Gaussian1D` object to the spectrum object created using `specutils`'s `Spectrum1D` class. At this stage, we found that we obtained some spurious measurements due to outliers in the spectra. Based on the expected values of the parameters of the Gaussian fit, apply another set of criteria (Equations 6–8) to eliminate false detections:

$$\frac{\text{Amp}_T^X(\mathcal{G}(\mathcal{L}_t^s))}{\sigma(\text{Amp}_T^X(\mathcal{B}_P))} \geq \min(5, \text{med} + 5\sigma), \quad (6)$$

$$\mu(\mathcal{G}(\mathcal{L}_t^s)) - E_T^X \in [-0.05, 0.05], \quad (7)$$

$$\sigma(\mathcal{G}(\mathcal{L}_t^s)) \in [0.05, 0.1]. \quad (8)$$

Here, $\text{Amp}_T^X(\cdot)$ denotes the amplitude, making Equation 6 the amplitude analog of the counts-based Equation 5. Equation 7 requires the Gaussian fit mean μ to be close to the known line energy E_T^X , and Equation 8 requires the line width to be consistent with expectations.

Figure 2 illustrates the steps described here. There are 238,817 XRF line detections that pass our set of criteria (Equations 6–8) from November 2019 to May 2025. As solar activity increases, there are more detections, especially for elements with higher atomic numbers. After processing the data, we observed that the signal-to-noise ratio was low for some of the line ratios. Hence, calculations for Mg/Si, Fe_L/Si, O/Si, and Ca-K α lines were repeated using 296 s binning.

2.3. Propagation of uncertainties

In the observed counts, we have contributions from the background and the source. By using propagation of errors we can obtain the variance in the background subtracted observed counts. We also assume poisson statistics for both the background and source photons for our uncertainty calculations.

Suppose that there are $N_{\mathcal{L}}$ spectra to be added to create the binned light spectra at time t ($\mathcal{L}_t$), and $N_{\mathcal{B}}$ spectra that are added to make up the background spectrum ($\mathcal{B}$). Then, by adding up the variances, scaled by the number of observations ($N_{\mathcal{L}}, N_{\mathcal{B}}$), of ($\mathcal{L}_t$) and ($\mathcal{B}$), we get:

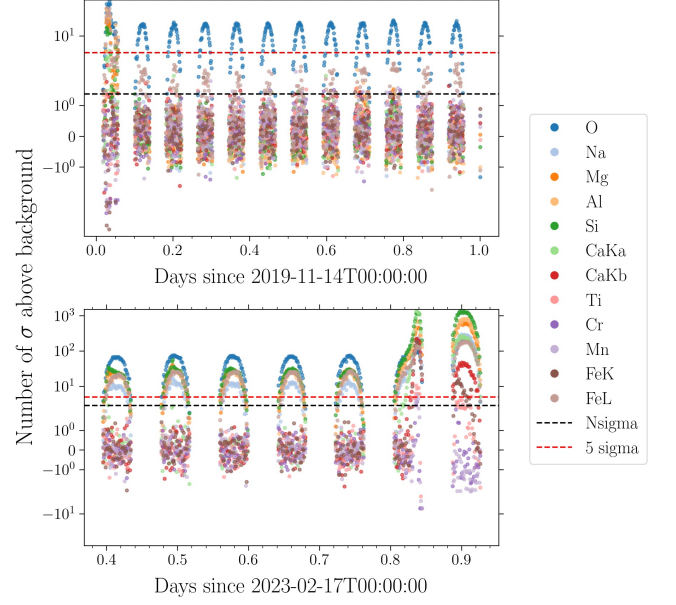


Figure 3. Light curves during a typical day during low solar activity (top) and high solar activity (bottom).

$$\text{std}(\mathcal{L}_t^s) = \sqrt{\frac{\mathcal{L}_t}{N_{\mathcal{L}}} + \frac{\mathcal{B}}{N_{\mathcal{B}}}} \quad (9)$$

We pass $\text{std}(\mathcal{L}_t^s)$ obtained from Equation 9 to the `Spectrum1D` object, which then weighs each data point accordingly. These weights are used to decide the optimal fit and are propagated in the fitted parameters.

3. ELEMENTAL RATIO MAPS

The maps are produced as a GeoTIFF file using the standard coordinate reference system (CRS) EPSG:4326. For calculating the ratios, we filter out the detections based on the criteria described in Equations 6–8, and calculate the area under the Gaussian fits for various elements. We define the ratio $\mathcal{R}_L^M$ of element M with element L as:

$$\mathcal{R}_L^M = \frac{A_M \sigma_M}{A_L \sigma_L}, \quad (10)$$

where A_M , A_L and σ_M , σ_L are the amplitudes and standard deviations of the Gaussian fits to the M and L spectral lines respectively. We take ratios with the base element as Silicon (or Aluminum in the case of Mg/Al) and derive the uncertainties from the covariance matrix given by `fit_lines` of the gaussian fits. We take the uncertainty weighted average and propagate uncertainties for each pixel. Each pixel in the map corresponds to an area of $\sim 5.3 \times 5.3$ km on the lunar surface.

Figure 4 shows the Mg/Al, Al/Si, Mg/Si, Fe_L/Si and Ca/Si maps which have global coverage. Mg/Al

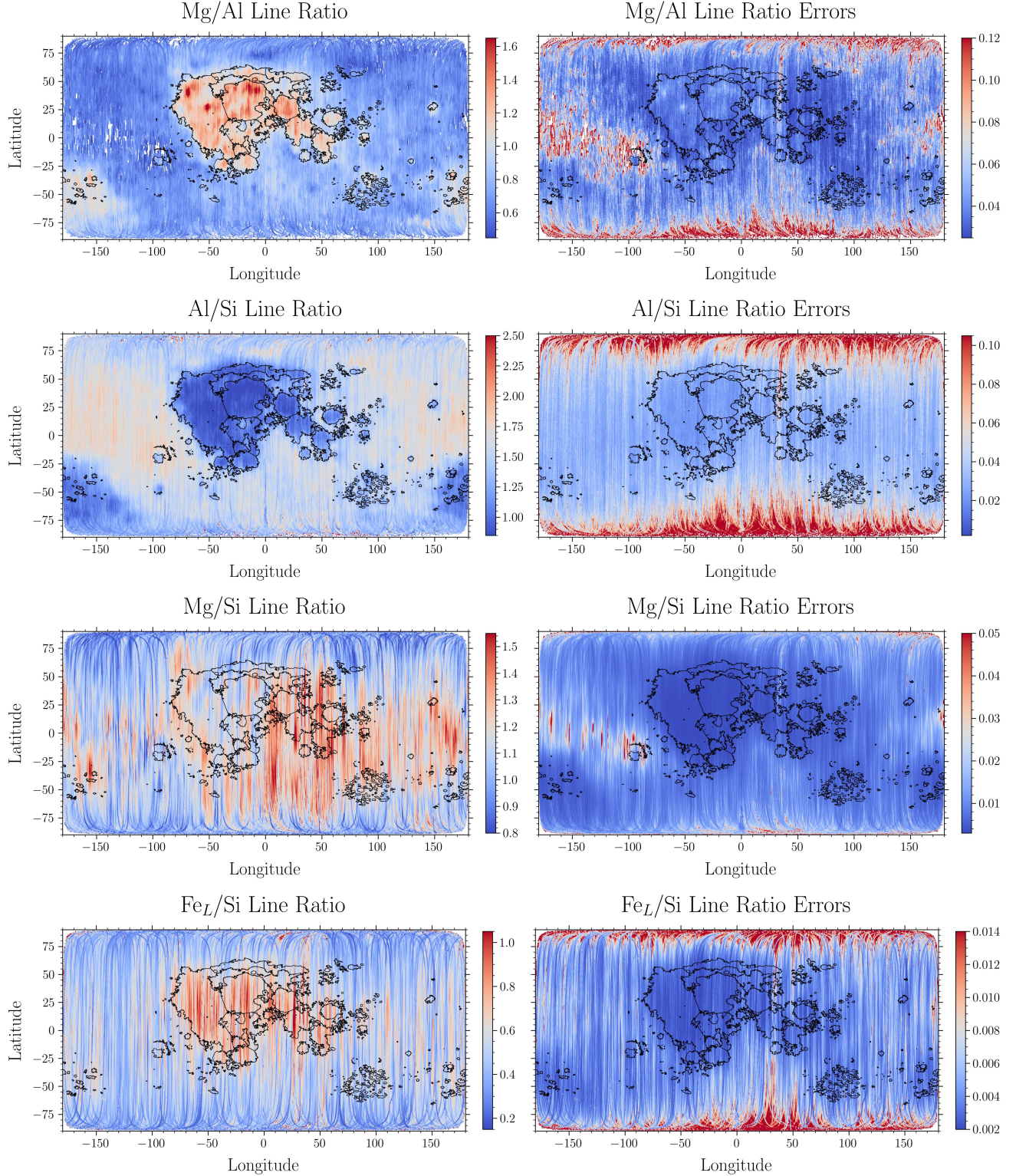


Figure 4. Lunar XRF line intensity ratio maps from CLASS data. Mg/Al and Al/Si best represents the distinction between the major lunar terranes. High Mg/Si values are not restricted to Mare while Fe_L/Si is higher in mare but not so evidently in SPA. Black lines are Mare boundaries from LROC data. As mentioned in §2.2, Mg/Al and Al/Si maps were created using 96 s data bins, while the Mg/Si and Fe_L/Si maps used 296 s binning.

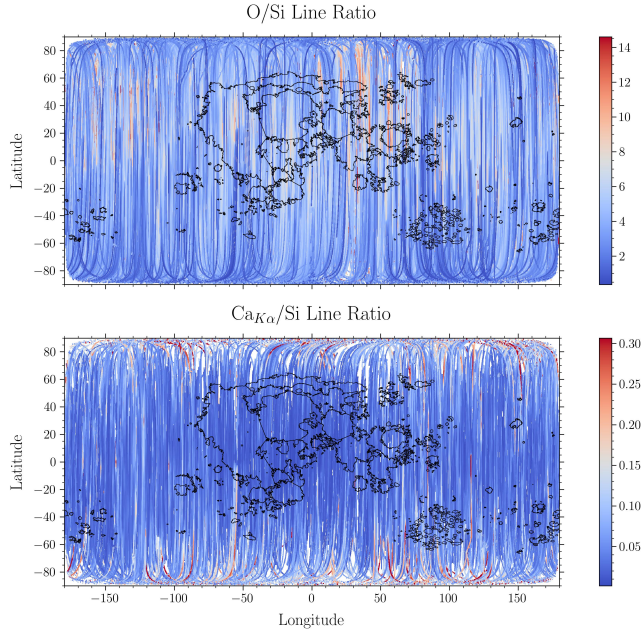


Figure 5. Lunar XRF line intensity ratio maps of O/Si and Ca-K α /Si from CLASS data binned at 296 s. Black lines are Mare boundaries from LROC data. Oxygen is not expected to vary much across the lunar surface, which is indeed observed in the O/Si map. On the other hand, the expected difference in terranes are not observed in Ca-K α map.

and Al/Si data best represent the known distinction in chemistry between Mare, highlands, and the South Pole Aitken Terrane (SPAT). The uncertainties are higher in the polar regions as the signal is weaker there because of the grazing solar incidence angles. For Mg/Al and Mg/Si, uncertainties are also higher in the highland terrane.

As a visual guide, we have added Mare boundaries on all maps. These boundaries are based on the digitization of lunar mare boundaries from Lunar Reconnaissance Orbiter Camera (LROC) Wide Angle Camera (WAC) basemap Shapefile, and downloaded from the Lunar Reconnaissance Orbiter Camera Science Operations Center website⁴.

O/Si and Ca-K α /Si maps shown in Figure 5 do not distinguish between the terranes. Oxygen is expected to be in the narrow abundance range of 40-45 wt % across the Moon, but it is difficult to validate the O/Si map, as there have been no such global measurements previously. The difference in XRF line energies between Si and Ca, along with the high line energy of Ca results in a larger dependence on the incident solar spectra.

⁴ https://data.lroc.im-ldi.com/lroc/view_rdr/SHAPEFILE_LROC_GLOBALMARE.

This dependence cannot be meaningfully mitigated by taking line intensity ratios (further discussed in §3.1), making the Ca/Si line ratio not suitable for representing composition. Na/Si, Ti/Si, Cr/Si and Fe-K α /Si have lower coverage and are not discussed here.

3.1. Methodological significance and limitations

A primary limitation of the current approach is the omission of the incident solar X-ray spectrum in the derivation of line ratios. As shown by Nittler et al. (2001), the ratio of characteristic XRF lines (e.g., Mg/Si or Al/Si) can vary by a factor of two or more depending on the solar coronal temperature. This dependence is more pronounced for heavier elements like Ca and Fe, which require higher-energy flare activity for excitation. By focusing our analysis on the soft X-ray range (0.5–2 keV) and adjacent elements like Mg and Al, we partially mitigate these effects, as the relative excitation of these lines remains more stable across varying solar conditions than their absolute intensities. The scattered solar spectrum also contains coronal line emission, which is not negligible at the lower end of our energy range. The Fe_L line is the most affected, as solar Fe-L emission lies at nearly the same energy as the lunar Fe-L fluorescence and is therefore not mitigated by taking a ratio. We accordingly treat the Fe_L/Si and Mg# maps as relative indicators of composition.

A second limitation is that we do not deconvolve the spectra with the detector response matrix. Since the response is energy dependent and no correction is applied, the measured ratios carry a multiplicative offset that is common to all pixels, affecting the absolute values but not the relative spatial variations used here.

Standard XRF analysis requires high-quality, concurrent solar spectral measurements to perform forward modeling. By utilizing an empirical line-ratio method, we can process a significantly larger fraction of the five-year CLASS dataset, including observations where solar monitoring data may be unavailable or difficult to reconcile. This approach enables the generation of high-resolution global maps that provide an approximate yet robust picture of lunar geochemistry, effectively resolving major terranes and geochemical dichotomies.

3.2. Correlation with abundances

X-ray line flux ratios can serve as an effective empirical tool for the initial probing of planetary bodies. For objects where ground truths are scarce or concurrent solar monitoring is unavailable, these uncorrected flux ratios allow for rapid, wide-scale terrain mapping to identify geochemically distinct regions of interest for subsequent detailed modeling.

To evaluate the utility of our empirical line-ratio approach, we compared our derived Mg/Al and Al/Si line intensity ratios with the corresponding modeled abundance (wt%) ratios from published CLASS maps in Narendranath et al. (2024) by cross matching each pixel from both sets of maps. To further contextualize these results, we extracted abundance ratios corresponding to sample return sites and compared them with the established ground-truth abundances of returned soils from Apollo, Luna (Meyer 2005), Chang'E 3 (Ng et al. 2023), Chang'E 5 (Zong et al. 2022), Chandrayaan-3 APXS (Vadawale et al. 2024), and Chang'E 6 (Li et al. 2024). We obtain the XRF line ratios for the abundance ratios corresponding to each of the sample return sites by taking the median of the line ratios in a 50 km square centered on the return site.

Figure 6 shows a positive trend between the measured line ratios and the fully modeled abundance ratios. We find the returned samples and the line ratios show a good correlation (with the exception of some outliers, namely Chang'E 5, Apollo 11 and Luna 24 in the Mg/Al plot). The abundance ratios from Narendranath et al. (2024) show noticeable scatter for both Mg/Al and Al/Si, along with an offset from the returned samples in the Al/Si plot. This is likely due to the published Si wt% maps from CLASS having a larger range of values (and higher uncertainty), along with a high spatial resolution ($\sim 5.3 \text{ km}^2$) compared to the 50 km^2 median value taken for the returned samples.

Despite these systematic offsets, the relative trends remain robust enough to differentiate major geochemical units. Mg/Al line ratio values $\mathcal{R}_{\text{Al}}^{\text{Mg}} \lesssim 0.8$ generally indicate highland regions, while for Al/Si, non-mare regions typically show $\mathcal{R}_{\text{Si}}^{\text{Al}} \gtrsim 1.25$.

We can derive the following first-order empirical relations to approximate the conversion of line ratios to weight-percent ratios:

$$\frac{\text{CLASS Mg wt\%}}{\text{CLASS Al wt\%}} = 0.88 \times \mathcal{R}_{\text{Al}}^{\text{Mg}} - 0.34 \quad (11)$$

$$\frac{\text{Sample Mg wt\%}}{\text{Sample Al wt\%}} = 0.91 \times \mathcal{R}_{\text{Al}}^{\text{Mg}} - 0.34 \quad (12)$$

$$\frac{\text{CLASS Al wt\%}}{\text{CLASS Si wt\%}} = 0.50 \times \mathcal{R}_{\text{Si}}^{\text{Al}} + 0.09 \quad (13)$$

$$\frac{\text{Sample Al wt\%}}{\text{Sample Si wt\%}} = 0.53 \times \mathcal{R}_{\text{Si}}^{\text{Al}} - 0.19 \quad (14)$$

The conversion enables better utilization of the global high-resolution maps derived here.

4. LUNAR CHEMISTRY WITH ELEMENTAL RATIOS

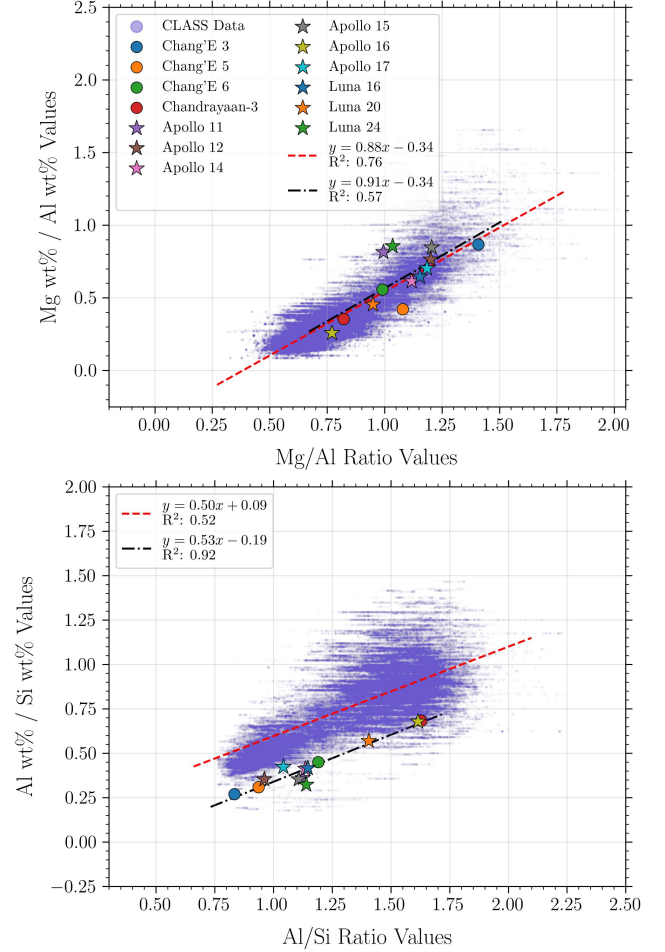


Figure 6. Comparing XRF line intensity ratios (this work) with elemental abundance data. Purple points denote inferred elemental weight ratios from Narendranath et al. (2024). The star symbols denote actual measurements from various missions. We see a positive linear correlation in the measurements. The dashed lines denote linear regression 1-D polynomial fits. The red dashed lines denote the fit to CLASS data, while the black dash-dotted line denotes the fit to elemental weight ratio data from various missions.

We hypothesize that the continuous multi-modal distribution observed in the data arises from underlying, independent populations corresponding to geographically diverse lunar terranes. By applying Gaussian Mixture Models (GMMs), we seek to capture and isolate these distinct geographic regions purely through their separation in compositional space.

Mineralogically, these statistically defined populations are intended to correspond to the dominant lunar surface materials, primarily mapping the transition between basaltic mineral suites (which characterize the lunar maria and the South Pole-Aitken basin) and

anorthositic mineral suites (which dominate the feldspathic highlands).

4.1. Clustering with Gaussian Mixture Models

To quantify the clustering observed in the elemental line ratio data, we applied an N -component Gaussian Mixture Model to the two-dimensional line ratio distributions (e.g., $\mathcal{R}_{\text{Si}}^{\text{Mg}}$ versus $\mathcal{R}_{\text{Si}}^{\text{Al}}$). Mathematically, the GMM represents the overall probability density of an observed data point as a weighted linear combination of N component multivariate Gaussian distributions (Reynolds 2009). Based on the GMM model, every data point is assigned a posterior probability of belonging to each of the N components, which is then used to assign the color mapping projected onto the lunar surface.

While the Bayesian Information Criterion identifies $N = 2$ as the optimal number of components for our data, this represents the primary binary dichotomy between the lunar highlands and mare. To better resolve the Moon's geochemical complexity, we also evaluate $N = 3$ and $N = 5$ configurations, which can potentially indicate transitional mixing zones. $N = 3$ is the highest component count for which the clusters correspond to distinct geological units, while $N = 5$ is included to illustrate the continuity of the transitions between them. The most powerful combination of ratios, according to our analysis, is the 2D distribution of $\mathcal{R}_{\text{Si}}^{\text{Mg}}$ and $\mathcal{R}_{\text{Si}}^{\text{Al}}$ (Figure 7, bottom panel) which easily bifurcates into a region of high Al and low Al, as expected.

The high Al/Si data points ($\mathcal{R}_{\text{Si}}^{\text{Al}} \gtrsim 1.2$) also show two distinct clusters (blue and green), which correspond to selenographically different regions (top panel, Figure 8). Plotting on the lunar mercator projection with this same color scheme, we can see three regions separated out (1) The Basaltic Lunar Maria (2) South Pole Aitken (SPA) Basin (3) the Feldspathic Highlands terrane. We apply a 5-component GMM which further distinguishes a low Mg/Si cluster in the north west highland Terrane and a low Mg/Si cluster in the north east (Figure 8, middle panel). Both models show the intermediate Mg/Si material that outlines the mare and the SPA.

We define an analogous quantity to the Magnesium Number (Mg#) as $\mathcal{R}_{\text{Si}}^{\text{Mg}} / (R_{\text{Si}}^{\text{Mg}} + R_{\text{Si}}^{\text{Fe}})$ and apply a 5-component GMM (Figure 8, bottom panel). Lower Mg# ratios appear in the mare and higher in highland terrane with a transition in between, as observed in previously published Mg# maps (Wang et al. 2021; Zhang et al. 2023). The large basins on the near side belong to the lowest Mg# cluster suggesting younger lava emplacements. In particular, isolated mare, (Moscoviense, Australe, Marginis, and Smythii) are grouped separately

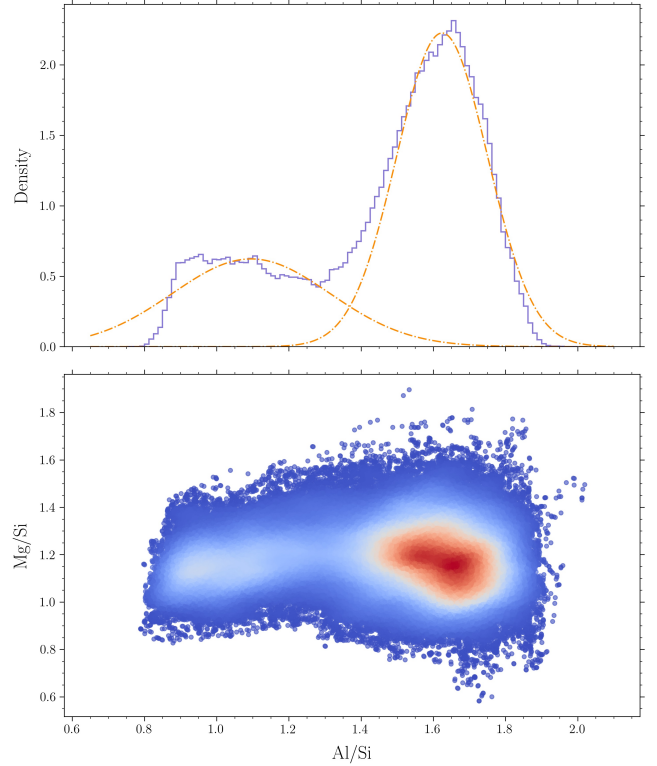


Figure 7. *Top panel:* A histogram of Al/Si line ratios (purple) measured over the entire surface shows a clear bimodal distribution, corresponding to the Mare and highland terranes. The Gaussian fits to this distribution are shown by the dash-dot orange line. *Bottom panel:* A two-dimensional distribution of the Mg/Si and Al/Si line ratios also shows a multi-modal distribution. The colormap shows the density of data points in the 2D space, going from low (blue) to high (red) density.

from the larger nearside basins. This grouping follows from their similar Al/Si values and does not imply that they are geochemically similar (Figure 4).

4.2. Global maps

We have generated global high-resolution maps of major elements with XRF line intensity ratios from CLASS on Chandrayaan-2. The Mg/Al and Al/Si maps are particularly useful to identify geochemically distinct regions that can be used for further detailed studies. The Mg/Al line intensity map is plotted with the contours of the mare basalts, along with notable geography commonly referenced in Figure 9.

The highest Mg/Al regions are associated with the near-side mare basins. Notable regions are the Aristarchus plateau and a region close to the boundary of Oceanus Procellarum towards the north east. Although the anomalously high Mg/Al ratio (≈ 2 compared to the global average of 0.75) north west of the

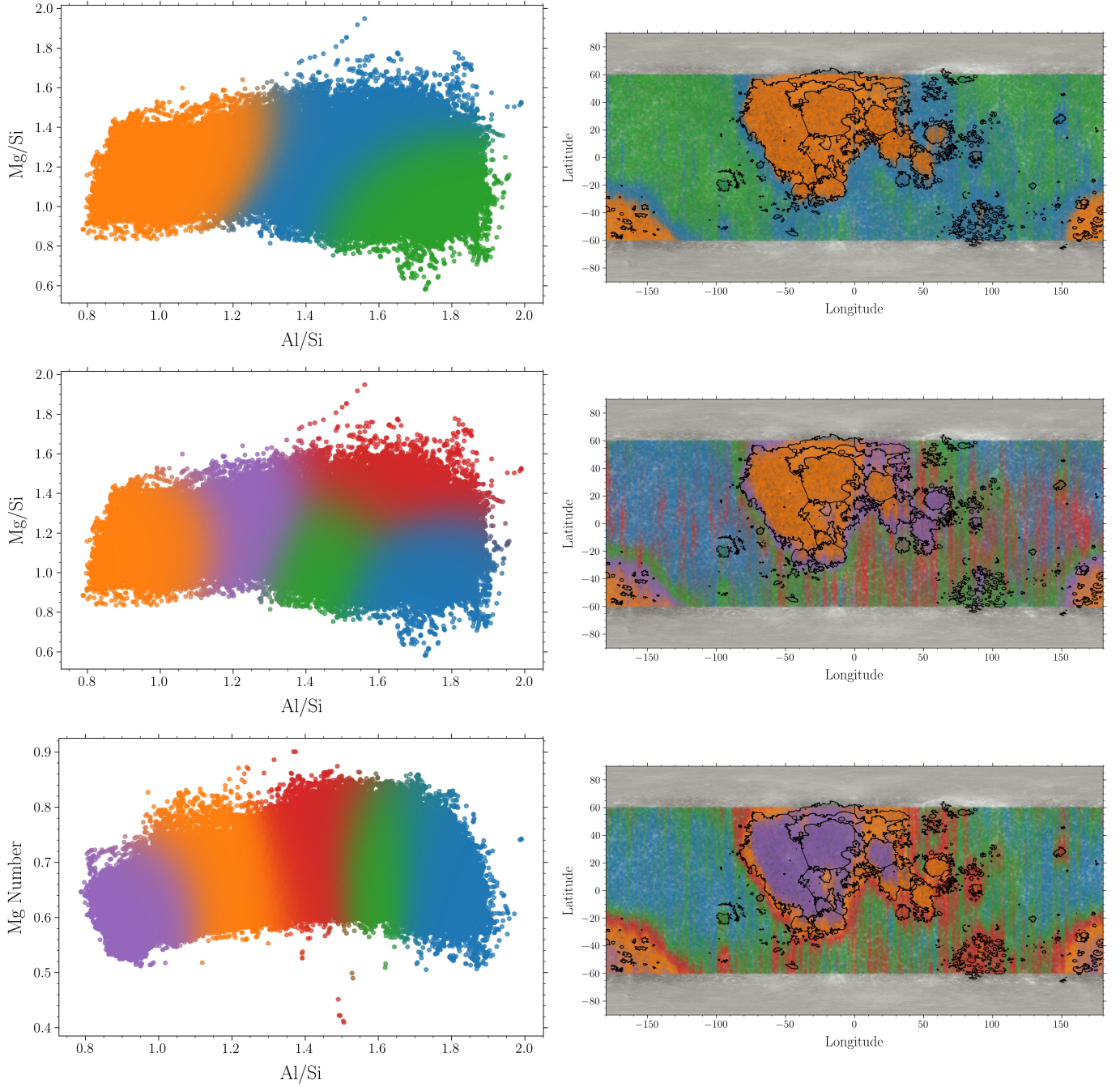


Figure 8. Lunar geochemical classification using Gaussian Mixture Models (GMM) on CLASS XRF line intensity ratios. In all maps, black contours represent mare boundaries from LROC data. Left panels show the 2D density distributions and GMM cluster assignments; right panels show the corresponding spatial distribution on a global Mercator projection. *Top row:* 3-component GMM applied to $\mathcal{R}_{\text{Si}}^{\text{Al}}$ vs. $\mathcal{R}_{\text{Si}}^{\text{Mg}}$ identifying the primary dichotomy between the Basaltic Maria (orange), and the Feldspathic Highlands (green), along with some transitional regions (blue) outlining the Maria and the South Pole Aitken basin. *Middle row:* 5-component GMM of $\mathcal{R}_{\text{Si}}^{\text{Al}}$ vs. $\mathcal{R}_{\text{Si}}^{\text{Mg}}$, further resolving geochemical sub-units and mixing zones. *Bottom row:* 5-component GMM applied to the distribution of the Mg# proxy derived from $(\mathcal{R}_{\text{Si}}^{\text{Mg}} / [\mathcal{R}_{\text{Si}}^{\text{Mg}} + \mathcal{R}_{\text{Si}}^{\text{Fe}}])$, grouping Mare Moscoviense, Australe, Marginis, and Smythii separately from the larger nearside basins.

Region	Al/Si	Mg/Si	Mg/Al
North Polar	1.48 ± 0.31	0.99 ± 0.11	0.64 ± 0.09
South Polar	1.36 ± 0.36	1.05 ± 0.12	0.73 ± 0.12
Highlands	1.59 ± 0.20	1.14 ± 0.14	0.69 ± 0.11
Mare	1.18 ± 0.17	1.12 ± 0.14	0.96 ± 0.22

Table 1. Average values of elemental line ratios in the lunar polar regions ($\pm 70^\circ$ to $\pm 90^\circ$), Highlands and Mare.

Site	Ratio	Apollo	This Work	This/Apollo
Mare Tranquillitatis	Al/Si	0.52 ± 0.10	1.00 ± 0.03	1.92 ± 0.37
	Mg/Si	0.62 ± 0.15	1.11 ± 0.05	1.79 ± 0.44
	Mg/Al	1.15 ± 0.24	1.10 ± 0.03	0.96 ± 0.20
Mare Fecunditatis	Al/Si	0.58 ± 0.13	1.10 ± 0.04	1.90 ± 0.43
	Mg/Si	0.51 ± 0.14	1.20 ± 0.02	2.35 ± 0.65
	Mg/Al	0.89 ± 0.29	1.12 ± 0.03	1.26 ± 0.41
Hadley-Apennines	Al/Si	0.53 ± 0.19	1.09 ± 0.03	2.06 ± 0.74
	Mg/Si	0.64 ± 0.20	1.31 ± 0.05	2.05 ± 0.64
	Mg/Al	1.26 ± 0.30	1.18 ± 0.01	0.94 ± 0.22
Descartes Highlands	Al/Si	0.81 ± 0.11	1.57 ± 0.09	1.94 ± 0.29
	Mg/Si	0.40 ± 0.14	1.32 ± 0.03	3.30 ± 1.16
	Mg/Al	0.49 ± 0.20	0.79 ± 0.01	1.61 ± 0.66
Taurus-Littrow	Al/Si	0.55 ± 0.09	1.12 ± 0.03	2.04 ± 0.34
	Mg/Si	0.55 ± 0.12	1.15 ± 0.07	2.09 ± 0.47
	Mg/Al	1.01 ± 0.26	1.06 ± 0.04	1.05 ± 0.27

Table 2. Comparison of Al/Si, Mg/Si, and Mg/Al ratios between Apollo XRFS measurements and this work at the five sites mentioned in Table 3 of Gloudemans et al. (2021). The last column gives the ratio of our values to the Apollo values, with propagated uncertainties. The weighted means across the five sites are 1.96 ± 0.17 for Al/Si, 2.08 ± 0.26 for Mg/Si and 1.02 ± 0.12 for Mg/Al.

Aristarchus crater coincides with high olivine abundances on the map derived from the Kaguya multiband imager (Lemelin et al. 2019), other similar regions do not. These regions are likely to be distinct in mafic content and composition, which requires detailed studies.

The maps in the polar regions (70° - 90°) are also shown in the lower panel of Figure 9. South Polar region has an average Mg/Al value of 0.73 ± 0.12 while it is 0.64 ± 0.09 for the North Polar region. In the 85° - 90° region where most of the future landing zones are located, the Mg/Al, Al/Si and Mg/Si ratios are similar for the polar regions.

Table 1 gives the average values of Mg/Si, Al/Si and Mg/Al ratios for the polar regions, highlands and mare. As seen from Table 2, our values for the Al/Si and Mg/Si ratios are consistently higher than those derived from Apollo XRFS (Gloudemans et al. 2021). The offset is a constant factor across the five sites, with weighted means of 1.96 ± 0.17 for Al/Si and 2.08 ± 0.26 for Mg/Si, rather than a terrane dependent difference. It likely arises because the CLASS quantum efficiency is reduced at the Si-K α line at 1.74 keV, which falls between the Al K-edge of the optical blocking filter at 1.56 keV and the Si K-edge of the detector at 1.84 keV (Pillai et al. 2021, Figure 8), underestimating the Si line flux. The gas proportional counters used in the Apollo missions are free of this effect. Mg/Al, for which both line energies lie below the Al K-edge, is 1.02 ± 0.12 and shows no such offset.

5. SUMMARY

We have developed a processing methodology to detect XRF signals from the CLASS calibrated spectral files and generate high-resolution maps of line intensity ratios from five years of data. Unlike conventional analyses that rely on forward modeling of incident solar spectra to derive absolute abundances, our approach utilizes raw XRF line intensity ratios as a proxy for surface composition. This provides a useful first order picture of the global distribution of lunar elements. Since the method requires neither concurrent solar measurements nor spectra with sufficient statistics for a full spectral fit, a much larger fraction of the observations becomes usable, allowing these maps to be generated at a finer spatial scale. The Mg/Al and Al/Si line intensity ratio maps show positive correlations with abundance ratios in wt% from return samples, as well as published CLASS elemental maps. The Mg/Al maps best represent the geochemical diversity that shows distinct regions that would be of interest for further, detailed studies.

Facilities: Chandrayaan-2

Software: Python v3.10.15 (Van Rossum & Drake 2009), NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), Astropy (Astropy Collaboration et al. 2013, 2018, 2022), Pandas (McKinney et al. 2010), Matplotlib (Hunter 2007)

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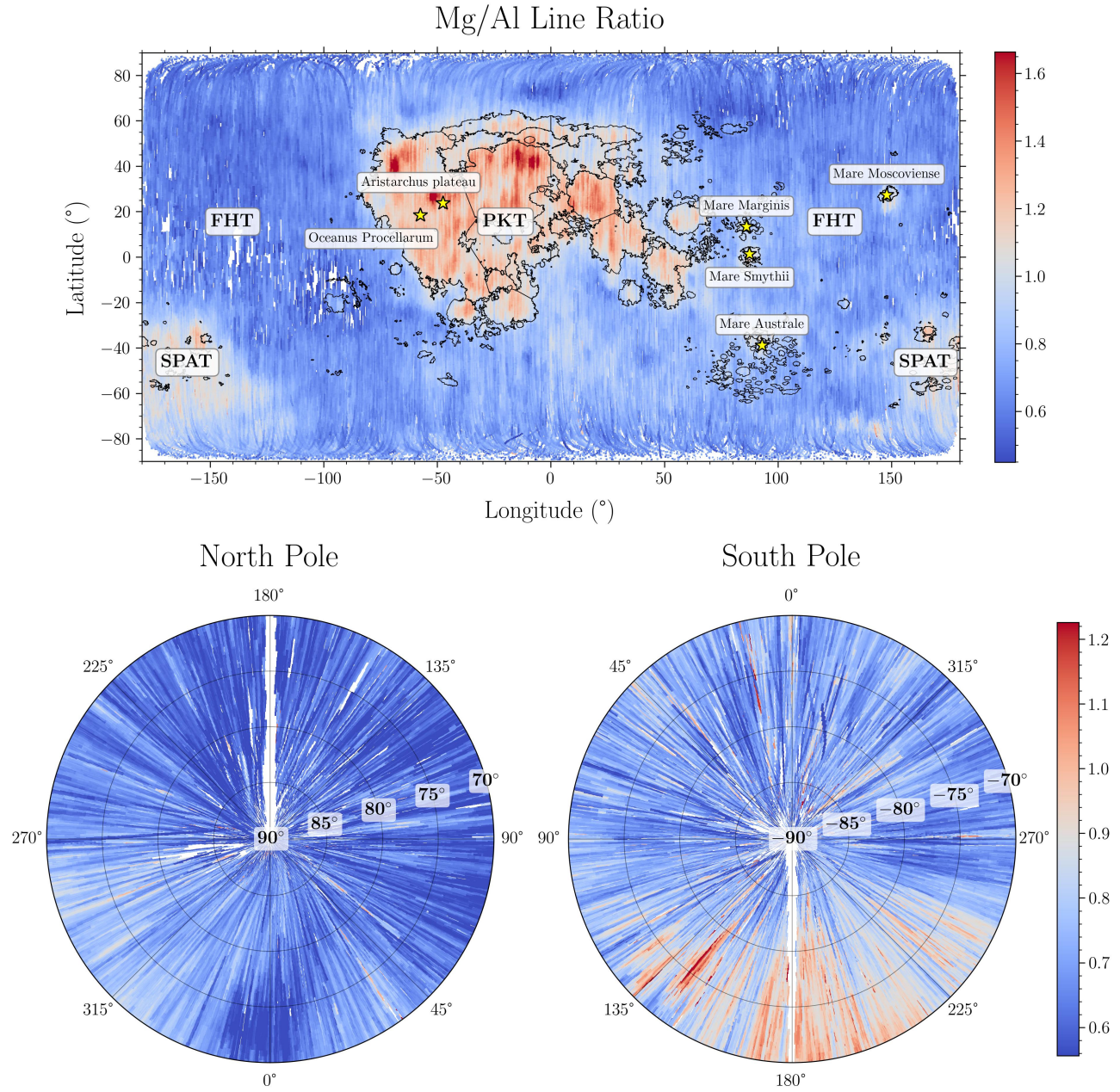


Figure 9. Lunar Mg/Al X-ray intensity maps. *Top panel:* Cylindrical projection. Black lines are Mare boundaries from LROC data. We see a clear distinction in the Mg/Al ratio between Mare and highlands. *Bottom panels:* Maps of the polar regions. The numbers around the periphery are longitudes, while the bold numbers denote latitudes.

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